International Year One Engineering

Computer Programming in C

Text in this font is the description of the task.

```
Text in this font is sample output of a program to the screen.
Example keyboard input to the screen appears as underlined e.g. 34
```

Exercises 4. Loops

1. Write a `for` loop to perform each of the following

   a. print all the even numbers between 0 and 100
   b. print all the odd numbers between 0 and 100.
   c. print the countdown sequence 10,9,8,7,6,5,4,3,2,1
   d. print all the numbers from 100 down to -100 in steps of 20.
   e. print all the numbers from -0.0375 to a maximum of 1.45 in intervals of 0.1225

2. Repeat Question 1 using `while` loops.

3. Repeat Question 1 using do-`while` loops.

4. Write a loop that keeps requesting a user to input the characters 'y' or 'n' in response to a question until they do so.

```
Do you want to continue (y/n)?: m
Response not recognized, try again.
Do you want to continue (y/n)?: b
Response not recognized, try again.
Do you want to continue (y/n)?: y

Thank-you, continuing program...
```

5. Write a program to print out a teaching multiplication table (up to multiplication by 12) for any user specified integer number between 1 and 12

```
Enter times-table to generate : 5

Multiplication table is:
1 x 5 = 5
2 x 5 = 10
3 x 5 = 15
:    :    :
11 x 5 = 55
12 x 5 = 60
```

6. Write code to evaluate the factorial of an integer n where $n!$ is defined for $n \geq 0$ by

$$n! = n(n-1)(n-2)\ldots(3)(2)(1)$$
$$\text{e.g. } 5! = 5(4)(3)(2)(1) = 5*4*3*2*1 = 120$$

Note that $0!$ is defined as equal to 1

```
Which number do you wish to find the factorial of ? 5

5! = 120
```

7. An infinite loop can be created by omitting all the expressions from a `for` statement i.e. simply `for(;;)`. The loop can only be terminated by a `break` statement from within the loop based on a conditional expression within the loop

```
for (;;)
{

      if ( …… ) break; /*breaks from infinite loop*/

}
printf("Loop terminated");
```

Using the above construction, write a program that repeatedly allows a user to input an integer and prints out its cubed value until the value 0 is entered.

```
Enter a number (enter 0 to stop): 3
3 cubed is 27
Enter a number (enter 0 to stop): 5
5 cubed is 125
Enter a number (enter 0 to stop): 0
Bye for now!
```

8. Repeat Question 7 using a while loop.

Projects 4. Loops

1. Write a program to print out all the prime numbers less than 100.

   (Hint: think about the definition of a prime number and then ask how you would test if a number is not a prime number)

2. Write a program that requests a user to input an integer number and then computes the sum of the digits of the number.

   ```
   Enter an integer number: 2358

   The sum of the digits is : 18
   ```

   (Hint: use %10 and /10)

3. Write a program that uses a loop to read several numbers representing the percentage exam scores of a student, then calculates the average and displays the result.

   Make sure the numbers entered by the user are non-negative and less than 100, and that the user has the freedom to input as many exam marks as desired.

   ```
   Enter percentage score : 74

   Do you wish to enter another score (y/n) ? : y

   Enter percentage score : -15

   Positive scores less than 100 only please!

   Enter percentage score : 58

   Do you wish to enter another score (y/n) ? : n

   Average score is : 66
   ```

4. The following Maclaurins series can be used to approximate the exponential function for a given value of x

   $$e^x \approx \sum_{n=0}^{N-1} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

   Write a program to evaluate the above series up to a user specified number of terms N :

   ```
   Enter a value of x to evaluate exp(x) : 0.5
   Enter number of terms of Maclaurins series : 5

   With 1 term  exp(0.5) series = 1
   With 2 terms exp(0.5) series = 1.500000
   With 3 terms exp(0.5) series = 1.625000
   With 4 terms exp(0.5) series = 1.645833
   With 5 terms exp(0.5) series = 1.645843
   ```

5. Write a program to print out a pattern of symbols \ and / to represent a horizontal arrow pattern as shown below. The user should indicate the width of the arrow as an integer number of characters

   ```
   Input width of arrow : 5

   \
    \
     \
      \
       \
       /
      /
     /
    /
   /
   ```

6. Write a program to print out a horizontal sinusoid pattern. Your program should request the time period of the waveform, given in terms of the number of vertical lines to print, and the peak amplitude of the waveform, given in terms of the number of horizontal characters.

```
Enter period of sine wave (no. of lines vertically) : 21
Enter amplitude of sine wave (no. of chars horizontally) : 20

                    *
                         *
                               *
                                   *
                                      *
                                       *
                                       *
                                     *
                                 *
                            *
                      *
                 *
           *
      *
  *
*
*
 *
    *
        *
              *
```